Benjamin A. Jasperson, Ph.D., P.E.

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EDUCATION

2024	**University of Illinois Urbana - Champaign**, Urbana, IL Ph.D. in Theoretical and Applied Mechanics NSF NRT Trainee (*DIGI-MAT*) w/ Graduate Concentration in Data Science & Engineering Dept. of Mechanical Science & Engineering Advisor: Harley T. Johnson, Ph.D. *Towards Data-Driven Inverse Design for Materials and Structures*
2010	**University of Wisconsin - Madison**, Madison, WI M.S. in Mechanical Engineering Dept. of Mechanical Engineering Advisors: Frank Pfefferkorn, Ph.D., and Kevin Turner, Ph.D. *Development and Calibration of Microscale Heat Flux Sensors Fabricated on Bulk Copper Substrates*
2008	**University of Wisconsin - Madison**, Madison, WI B.S. in Mechanical Engineering w/ Certificate in Business Dept. of Mechanical Engineering

PROFESSIONAL EXPERIENCE

2024 -	Postdoctoral Scholar - Research Associate Prof. Krishna Garikipati's group University of Southern California, Los Angeles, CA
2017 - 2020	Mechanical Engineer, Phoenix Nuclear Labs, Madison, WI *Focused on the design and production of high-yield fusion neutron sources for the medicine, defense, and energy sectors. Now a part of SHINE Technologies.*
2010 - 2017	Senior Mechanical Design Engineer, Plexus Corp., Neenah, WI *Concept, design, verification, and manufacturing transfer for devices in healthcare, industrial, defense/aerospace, and telecom.*

PUBLICATIONS

See *Google Scholar* for a research impact summary.

Journal Articles (Peer-Reviewed)

2026	B. L. Boyce, M. A. Wood, K. Garikipati, A. E. Robertson, J. Larson, I. Srivastava, B. Debusschere, S. Desai, P. Iyer, P. Robbe, M. Cherukara, T. Munson, M. Du, T. Mohanty, D. J. Gardner, L. Capolungo, **B. A. Jasperson**, J. Tan, R. Dingreville, "Materials Behavior as Mechanism Ensembles: A Probabilistic Framework for Emergent Behavior," under review, *Current Opinion in Solid State & Materials Science*, 2026. *arXiv*
2026	M. F. Shojaei*, R. Gulati*, **B. A. Jasperson***, S. Wang, S. Cimolato, D. Cao, M. Vardhan, W. Neiswanger, and K. Garikipati, "AI University: An LLM-Based Learning Assistant Aligned with Course-Specific Content in the Finite Element Method," under review, *Transactions on Machine Learning Research*, 2026. *arXiv* (* Equal Contribution)

2025	**B. A. Jasperson**, I. Nikiforov, B. Runnels, H. T. Johnson, and E. B. Tadmor, "Fundamental microscopic properties as predictors of large-scale quantities of interest: Validation through grain boundary energy trends," *Acta Materialia*, p. 120722, Jan. 2025. *DOI*
2025	**B. A. Jasperson**, I. Nikiforov, A. Samanta, F. Zhou, E. B. Tadmor, V. Lordi, and V. V. Bulatov, "Cross-scale covariance for material property prediction," *npj Computational Materials*, vol. 11, no. 1, p. 1, Jan. 2025. *DOI*
2024	**B. A. Jasperson** and H. T. Johnson, "A data-driven method for optimization of classical interatomic potentials," *MRS Advances*, vol. 9, no. 11, pp. 863–869, Mar. 2024. *DOI*
2024	**B. A. Jasperson**, M. G. Wood, and H. T. Johnson, "A dual neural network approach to topology optimization for thermal-electromagnetic device design," *Computer-Aided Design*, vol. 168, p. 103665, Mar. 2024. *DOI*
2014	**B. A. Jasperson**, J. Schmale, W. Qu, F. E. Pfefferkorn, and K. T. Turner, "Thin film heat flux sensors fabricated on copper substrates for thermal measurements in microfluidic environments," *J. Micromech. Microeng.*, vol. 24, no. 12, p. 125018, Dec. 2014. *DOI*
2010	**B. A. Jasperson**, Yongho Jeon, K. T. Turner, F. E. Pfefferkorn, and Weilin Qu, "Comparison of micro-pin-fin and microchannel heat sinks considering thermal-hydraulic performance and manufacturability," *IEEE Trans. Comp. Packag. Technol.*, vol. 33, no. 1, pp. 148–160, Mar. 2010. *DOI*

Peer-Reviewed Conference Proceedings

2026	**B. A. Jasperson**, K. Garikipati, S. Srivastava, "Resource Analysis for Quantum Simulation of Spatially Varying Transport–Reaction Equations," Workshop on QC for PDEs: Algorithms, Applications & Challenges, 2026 IEEE International Conference on Quantum Computing & Engineering (QCE), 2026. (accepted manuscript)
2021	M. Wood, A. McKay, T. Morin, D. Serkland, T. Luk, S. Wolfley, L. Gastian, J., **B. A. Jasperson**, and H. T. Johnson, "Optically-triggered optical limiters for short-wavelength infrared sensor protection," presented at CLEO, virtual, STh1E.3, 2021. *URL*
2009	C. Konishi, W. Qu, **B. A. Jasperson**, F. Pfefferkorn, and K. T. Turner, "Experimental study of adiabatic water liquid-vapor two-phase pressure drop across an array of staggered micro-pin-fins," *ASME International Mechanical Engineering Congress and Exposition Proceedings*, v10, p 1597-1605, 2009. *DOI*
2010	**B. A. Jasperson**, F. E. Pfefferkorn, W. Qu, and K. T. Turner, "A thin-film heat flux sensor fabricated on copper for heat transfer measurements in parallel channel heat sinks," *Proceedings of the 5th International Conf. on Micromanufacturing*, p 437-444, 2010. *URL*

Manuscripts in Preparation

2026	**B. A. Jasperson**, L. Dehay, R. Gulati, S. Srivastava, and K. Garikipati, "State-dependent spatiotemporal agent dynamics of potential-driven situational response"
2026	**B. A. Jasperson**, Q. T. Tran, I. D. Boureima, H. Mourad, and K. Garikipati, "A global-local approach to phase-field fracture mechanics using an integrated machine learning-PDE solver"
2026	I. Nikiforov, D. Karls, C. Waters, E. Fuemmeler, **B. A. Jasperson**, K. Sheriff, J. E. Ogbebor, R. Freitas, G. Zhang, P. Hoellmer, T. Egg, S. Martiniani, R. Elliott, N. Bernstein, Cyberloop Personnel, AFLOW Personnel, and E. B. Tadmor, "Crystal Genome: OpenKIM interatomic potential testing framework for arbitrary AFLOW crystal prototypes"

PATENTS

2019	P. Anderson, K. Novak, K. Mclennan, M. Mackaplow, G. Song, G. S. Dhami, **B. Jasperson**, S. Smieja, M. Svacina, "Devices and methods for delivering a beneficial agent to a user," 10213546, Feb. 26, 2019.

RESEARCH SUPPORT & RESOURCE ALLOCATIONS

Awarded Grants & Resource Allocations

2026	Allocation Manager & Lead Renewal Author, ACCESS *Project*: "Learning Mechanics-Driven Multiphysics Phenomena Using Generative AI on Spatiotemporal Simulation Data" *Allocation*: 1.5M ACCESS Credits (~28K GPU-hours)
2026	External Collaborator (PI: Lampros Svolos), UVM Scholarship of Teaching and Learning *Project*: "Teaching-Aligned AI Learning and Rubric-Guided Feedback in Engr. Mechanics" *Total Amount*: $8,500

Proposal Development & Grantsmanship

2026	Proposal support, Laboratory Directed Research & Development (LDRD) *Project*: GenIce: A Generative AI Framework for High Resolution Sea Ice Forecasting from Short-Term to Seasonal Scales *Status*: Submitted Jun 2026, Under Review
2026	Co-PI, U.S. Department of Energy (DOE BES) *Funding call*: The Genesis Mission: Transforming Science and Energy with AI *Project*: "Bridging Scientific Models and Hybrid Quantum-Classical Workflows through AI-Driven Unitary Composition" *Status*: Submitted April 2026, Under Review, Total Requested: $750,000
2026	Co-PI, ASCR Leadership Computing Challenge (ALCC) *Project*: "Microstructure Insights through Reliable/Interpretable AI and Guided Experiments" *Allocation*: 964.5K node-hours (Aurora and Frontier) *Status*: Submitted Jan 2026, Under Review

INVITED TALKS

2026	B. Jasperson, L. Dehay, J. Tan, S. Siddhartha, K. Garikipati, "Modeling of agent interactions in a complex landscape," Society of Engineering Science (SES) Annual Technical Meeting, West Lafayette, IN, Oct 11-14, 2026 (Accepted talk)
2024	"Towards data-driven inverse design for materials and structures" National Institute of Standards and Technology (NIST), Thermodynamics and Kinetics Group

RESEARCH / CONFERENCE PRESENTATIONS

Presentations Given

2026	**B. Jasperson**, D. Cao, P. Jain, J. Tan, W. Neiswanger, K. Garikipati, "On the accuracy and stability of spatiotemporal ViT solvers for mechanics," 20th U.S. National Congress on Theoretical and Applied Mechanics (USNC-TAM26), Pasadena, CA, June 21-25, 2026
2025	**B. Jasperson**, R. Gulati, Q. T. Tran, I. D. Boureima, H. Mourad, K. Garikipati, "Machine learning solvers for phase-field fracture mechanics," Society of Engineering Science (SES) Annual Technical Meeting, Atlanta, GA, Oct 12-15, 2025
2025	**B. Jasperson**, Q. T. Tran, I. D. Boureima, H. Mourad, K. Garikipati, "A global-local approach to phase-field fracture mechanics using an integrated machine learning-PDE solver," 18th U.S. National Congress on Computational Mechanics (USNCCM), Chicago, IL, July 20-24, 2025
2025	**B. Jasperson**, M. F. Shojaei, R. Gulati, S. Wang, S. Cimolato, D. Cao, W. Neiswanger, K. Garikipati, "Creating an AI Professor for a finite element methods course," Engineering Mechanics Institute Conference (EMI 2025), Anaheim, CA, May 27-30, 2025
2025	**B. Jasperson**, I. Nikiforov, A. Samanta, F. Zhou, B. Runnels, H. Johnson, V. Lordi, V. Bulatov, E. Tadmor, "Cross-scale covariance for material property prediction," Engineering Mechanics Institute Conference (EMI 2025), Anaheim, CA, May 27-30, 2025

2024	**B. Jasperson**, H. Johnson, "Towards data-driven inverse design for interatomic potentials," Engineering Mechanics Institute Conference and Probabilistic Mechanics & Reliability Conference (EMI/PMC 2024), Chicago, IL, May 27-30, 2024.
2023	**B. Jasperson**, H. Johnson, "Using data and machine learning to simplify and accelerate inverse design and model development in materials," poster, Society of Engineering Science (SES) Future Faculty Symposium, Minneapolis, MN, Oct 8-11, 2023.
2023	**B. Jasperson**, I. Nikiforov, H. Johnson, E. Tadmor, "Predicting grain boundary energy from few-atom simulations: A study across interatomic potentials," Society of Engineering Science (SES) Annual Technical Meeting, Minneapolis, MN, Oct 8-11, 2023.
2023	**B. Jasperson**, M. Wood, H. Johnson, "Inverse design and fabrication of a vanadium dioxide optical device using a dual neural network topology optimization approach," 17th U.S. National Congress on Computational Mechanics (USNCCM), Albuquerque, NM, July 23-27, 2023.
2022	**B. Jasperson**, M. Wood, H. Johnson, "Optimization of an optical shutter using machine learning," Society of Engineering Science (SES) Annual Technical Meeting, College Station, TX, Oct 16-19, 2022.
2022	**B. Jasperson**, "Optimization of an optical shutter using machine learning," Harnessing Data for Materials Symposium, Chicago, IL, Aug 19-30, 2022.
2010	**B. Jasperson**, F. Pfefferkorn, W. Qu, K. Turner, "A thin-film heat flux sensor fabricated on copper for heat transfer measurements in parallel channel heat sinks," 5th International Conference on Micromanufacturing (ICOMM), Madison, WI, April 5-8, 2010.

Coauthored Presentations (selected)

2025	E. Tadmor, **B. Jasperson**, I. Nikiforov, A. Samanta, F. Zhou, B. Runnels, H. Johnson, V. Lordi, V. Bulatov, "Cross-scale covariance for material property prediction," presented at APS Global Physics Summit, March 16-21, 2025.
2025	R. Gulati, **B. Jasperson**, K. Garikipati, "Transformer models in continuum mechanics," presented at SIAM Conference on Computational Science and Engineering, Fort Worth, TX, March 3-7, 2025
2009	B. Smith, **B. Jasperson**, and S. Manakasettharn, "Micro-machined molds for manufacturing micro-fluidic devices using soft lithography," poster, presented at International Manufacturing Science and Engineering Conference-MSEC, West Lafayette, IN, Oct 4-7, 2009.

CAMPUS / DEPARTMENTAL TALKS

2023	"Towards data-driven inverse design for materials and structures," seminar speaker, Virtual, iShare seminar series (UIUC, UIC, Duke), July 2023
2022	"Optimization of an optical shutter using machine learning," Sandia Academic Alliance - University of Illinois LDRD Mini-Conference, Urbana, IL, Sept 2022.
2022	"Experiences / lessons learned from post-grad school industry life," UIUC, DIGI-MAT Professional Development Seminar, July 2022
2021	"Rclone," UIUC, DIGI-MAT Professional Development Seminar, July 2021

RESEARCH EXPERIENCE

2025 -	Guest Scientist - T-3 Division Los Alamos National Laboratory (LANL)
2020 - 2024	Research Assistant, Prof. Harley Johnson's group University of Illinois Urbana-Champaign, Urbana, IL
2023	DIGI-MAT Graduate Internship, Prof. Ellad Tadmor's group University of Minnesota Twin Cities, Minneapolis, MN

2021 - 2023	National Science Foundation (NSF) NRT Trainee University of Illinois Urbana-Champaign, Urbana, IL
2008 - 2010	Research Assistant, Prof. Frank Pfefferkorn's and Prof. Kevin Turner's groups University of Wisconsin - Madison, Madison, WI

TEACHING AND MENTORING EXPERIENCE

2026	Instructor, USNC/TAM26 Short Course, "Foundation Models in Mechanics," June 2026
2026	Guest lecturer, Machine Learning and Computational Physics (AME 508), University of Southern California
2025	Instructor, USNCCM Short Course, "Fine-tuning large language models for Computational Mechanics," July 2025
2025	"Cross-scale covariance for material property prediction," MSE 598 guest lecturer, University of Illinois Urbana-Champaign, Spring 2025 (*link*)
2024 -	Mentor to multiple master's students, Prof. Krishna Garikipati's group, University of Southern California
2023	Teaching Assistant, Introductory Solid Mechanics (TAM251), University of Illinois Urbana-Champaign, Fall 2023
2022 - 2023	Mentor, Undergraduate Research Apprenticeship Program (URAP), University of Illinois Urbana-Champaign
2017 - 2020	"Project Planning for Engineers," ME 490 Senior Design guest lecturer, Milwaukee School of Engineering
2016 - 2018	"Prototyping," ME Senior Design guest lecturer, University of Wisconsin-Madison
2012	Mentor, FIRST Robotics, NEW Apple Corps - Team 93 (Appleton, WI)
2010 - 2020	Industry mentor for multiple interns and full-time hires

AWARDS, GRANTS AND ACHIEVEMENTS

2025	USNCCM18 Travel Award
2024	Top ten finalist, USNC/TAM 5MT Virtual Thesis Competition
2023	List of "Teachers Ranked as Excellent by Their Students" University of Illinois Urbana-Champaign, Fall 2023
2023	Accepted to the Future Faculty Symposium Society of Engineering Science, 2023
2023	Mavis Future Faculty Fellow (MF3) University of Illinois Urbana - Champaign
2021 - 2023	NSF Graduate Traineeship *DIGI-MAT*: Data and Informatics Graduate Intern-Traineeship: Materials at the Atomic Scale University of Illinois Urbana - Champaign
2014	Article selected as "Highlights of 2014", *J. Micromech. Microeng.*
2008	Graduated with Distinction, University of Wisconsin-Madison

LICENSES AND CERTIFICATIONS

2024	Certificate in Foundations of Teaching Center for Innovation in Teaching & Learning (CITL) University of Illinois Urbana-Champaign

2023	Graduate College Mentoring Certificate University of Illinois Urbana-Champaign
2016 -	Professional Engineer (P.E.), State of Wisconsin

ACADEMIC AND PROFESSIONAL SERVICE

2026	Session co-chair and organizer, "Foundation models for mechanics" Minisymposium, 20th U.S. National Congress of Theoretical and Applied Mechanics (USNC-TAM26), Pasadena, CA, June 21-25, 2026
2025 -	Journal Reviewer: Journal of Nonlinear Science
2024	Volunteer Judge, Undergraduate Research Symposium University of Illinois Urbana - Champaign
2021 - 2022	Students Advising on Graduate Education (SAGE) University of Illinois Urbana - Champaign
2021 -	USACM - Member
2019 -	ASME - Member
2005 -	Tau Beta Pi – Wisconsin Alpha Chapter, Illinois Alpha Chapter

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